Architecture & Trade-offs

Yazan Tommalieh · November 2026 · Companion to the Part 1 repository

The core has one primitive: `balance(account, value_date_upto, as_of_posting_day)`, a sum over an append-only log where every record carries the day it was posted and the day it is effective. There is no running balance; holds, fees and interest are all queries over the log. That is why it is correct under backdated entries, and why it is slow.

1 Append-only at scale

What breaks first: the end-of-day close

Every close re-checks every past day for every account, because a backdated entry can make a closed day negative, and each check scans the whole log. Cost per close is about *days × accounts × entries*, and entries grow with *days × accounts*: quadratic in both. At 100× accounts that is roughly 10,000× the work against a fixed deadline. Nothing goes wrong; it just stops finishing on time.

Where state grows without bound

The three logs (entries, authorizations, errors) grow by design: they are the audit record. The set of seen event ids grows by accident. Every lookup is a linear scan, and one global sequence number makes the engine a single writer.

> **Cheapest fix: close the period**
>
> Pick a horizon *H*, the last closed business day. Refuse entries value-dated before *H* (one comparison in `append_entry`) and, at each close, snapshot every account's balance, holds and fee flags as of *H*. A balance is then snapshot plus a scan of entries after *H*, and the close re-checks only the days since *H*. Audit is unchanged because the snapshot derives from the log. Still open afterwards: the log itself (five-year retention) and per-account sequences for parallelism.

2 Value-dated entries in production

A value-dated entry rewrites history. In Part 1 one backdated debit restated three closed days, produced three fees in one close and declined an authorization; its reversal restated the days again and left the fees standing. In a UAE-licensed bank each effect has an owner.

Customers and complaints	Issued statements become wrong and fees appear for days the customer saw as positive. CBUAE Consumer Protection Standards (Circular 8/2020, Art. 8): acknowledge a complaint in 2 business days, answer in 30, keep the file five years. Re-assessed fees will be the main source of those complaints. (EU PSD2 Art. 87(3) forbids a debit value date earlier than the debit itself; here, legitimate backdating is bank-side only: scheme clearing, corrections, adjustments.)
Fees	Overdraft charges sit under the CBUAE retail fee caps (Regulations 29/2011, Annexure 2); a new fee or a change above 5% needs prior approval. One backdated debit becoming three charges is a fee behaviour compliance must sign off and the customer agreement must describe.
Finance	Interest is recomputed from restated history, so a backdated entry across a month-end changes reported income and needs reversing journals the core does not produce.
Audit and AML	Both dates on every record make “what did we know on Day 5” a query and meet the five-year retention in Federal Decree-Law No. 10 of 2025. Missing: who backdated an entry, and why.

> **The one control before go-live: a backdating horizon with maker-checker**
>
> The core refuses entries value-dated before the last closed day. Anything that must land behind it goes into an adjustments queue; a second person approves it with a reason code, and the approval is appended to the log. This bounds how much history can be rewritten, gives the examiner the who-and-why trail, and makes automatic fee re-assessment safe. The second control, linking each fee to its trigger so reversing the cause reverses the fee, is the failing test in Part 1 (Fees, below).

3 Authorization lifecycle

An authorization is a decision record; its state (declined, active, settled) is derived from whether a settlement references it. The only ending implemented in full is a matching settlement. The others:

References: CBUAE Consumer Protection Standards (Circular 8/2020) Art. 8 · CBUAE Regulations 29/2011 Annexure 2 (2018) · Federal Decree-Law No. 10 of 2025 · PSD2 Art. 87 · Visa/Mastercard hold timeframes. Drafted with Claude, as disclosed in the Part 1 WORKLOG.

ENDING	REAL-WORLD CASE	WHAT I WOULD MANDATE
Declined at request today: recorded, never revisited	Insufficient available funds; in practice also fraud rules.	Final, with a reason code; not re-opened by a later backdated credit.
Duplicate request today: rejected as an error	Network retry.	Return the original decision; no second hold.
Never settled today: hold lives forever	Abandoned pre-authorization: hotel, car rental, fuel, cancelled order.	Expire per scheme rules (Visa 7 days, 31 for lodging and rental; Mastercard 7, 30 for pre-authorizations); release the hold, record the expiry.
Reversed by the merchant today: no such event	Order cancelled before capture.	New event: full reversal voids the hold, partial reduces it; merchant-originated only.
Late presentment / no authorization today: rejected (criterion 4)	Stand-in processing, offline chip approval, force-post.	Post it flagged so the ledger matches money the scheme already settled; route to disputes. The remedy is a chargeback, not refusal.
Settled for a different amount today: partial releases all; over flagged	Bill below the estimate; tips above it.	Partial releases the remainder unless multi-capture was flagged; over-capture within the scheme's tolerance for the merchant type.
Settlement later reversed today: funds return, no hold	Refund or chargeback.	Same as today: a refund is a new credit, a chargeback its own workflow.
Account frozen or closed today: not modelled	Fraud freeze, legal hold, dormancy, closure.	Freeze blocks new authorizations but keeps existing holds; closure releases them and routes late clearings to disputes.
Incremental authorization today: same id is a duplicate	Hotel stay extended.	Amend the existing hold and reset its expiry.

4 What I cut and why

STORAGE AND SCALE

In-memory lists, linear scans, one global sequence (persistence out of scope)
Quadratic close; single writer; a crash loses the idempotency set, so a replay can double-post.

Backdating unbounded (the exercise depends on it)
Full-history restatement; covered by the horizon in section 2.

MONEY MODEL

Single-sided entries, no double-entry, no transfers (none in the stream)
No debits-equal-credits check; nothing reconciles to the general ledger.

Simple daily interest, positive only, one capitalisation; no debit interest, tiers or Islamic profit model (spec wording)
Wrong for any real product; a restated accrual across a reporting period has no journal.

HALF_UP, rounded daily then summed (spec wording)
Must match the customer agreement; the difference is fils.

Only an AED fee configured (the spec gives one)
An overdrawn BHD account raises an error instead of a charge.

FEES

No link from a fee to its trigger, no fee reversal (spec never says when to unwind; the failing test)
A reversed bank error leaves the customer paying. Fix: a trigger reference on each derived entry and a compensating fee reversal.

Debits never declined, no overdraft limit, flat fee with no cap (spec gives none)
Unlimited overdraft and fee stacking; needs a limit per account and a cap per cycle.

AUTHORIZATIONS

No hold expiry, merchant reversal, incremental authorization or multi-capture (none in the stream)
Holds lock funds forever; a merchant void, the most common real ending, cannot be represented.

Approvals never revisited after a backdated debit (how card networks work)
Exposure sits with the bank; nothing flags “approved on a balance later restated”.

No account lifecycle (out of scope)
Freeze and closure behaviour for holds is undefined.

TIME

Integer days, no calendar (the brief's days are abstract)
Weekends, holidays, cut-offs and Gulf-time day boundaries are undefined.

Future value dates refused (keeps “balance as of day d” unambiguous)
Standing and scheduled payments cannot be represented.

INPUT AND ERRORS

Structural faults stop the replay (right for a test harness)
One bad message halts production; needs a dead-letter queue.

Rejections are only logged (the brief asks for errors to be printed)
A real force-post never reaches disputes; settled money goes unaccounted.

No operator, channel, reason code or reference on entries (none in the stream)
Audit knows what and when, not who or why; no reconciliation to scheme files.

In one line: a query over an immutable log, right and slow. Production keeps the log and adds a horizon, links on derived entries, and the authorization events the card world actually sends.

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